HARSHIT SHYAMSUKHA

SOFTWARE ENGINEER · AI/ML & FULL-STACK

Final-year CS student who takes projects past the demo stage — into the part where something holds up under real use. Work spans AI/ML systems (LLM fine-tuning, agentic pipelines, forecasting) and full-stack engineering (FastAPI, React/TypeScript, PostgreSQL). Debugged live memory and auth failures in production, ran full security audits fixing 12 vulnerabilities, and shipped a forecasting model that beat its baseline by 12.3%. Red Hat OpenShift certified. Comfortable across containerised, multi-service systems.

AI / ML · Full Stack · Agentic Systems · System Design · DevOps / Cloud

CONTACT

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TECHNICAL SKILLS

Languages
Python, TypeScript, JavaScript, C, C++, Java, SQL, HTML, Shell

Frameworks
FastAPI, React, Node.js, LangGraph, Docker, Git, PyTorch, Groq API, Ollama, HuggingFace (PEFT/LoRA/QLoRA), FAISS

Databases
PostgreSQL, Supabase, MongoDB, MySQL

Infrastructure
Kubernetes, Terraform, Ansible, AWS (VPC, EKS, RDS, S3, IAM), OpenShift

Core Concepts
DSA, DBMS, OS, System Design, OOP, Security Auditing

EDUCATION

SRM Institute of Science & Technology
B.Tech Computer Science
Chennai · 2023 – 2027

CERTIFICATIONS

Red Hat OpenShift Dev I (DO188)
Red Hat · Nov 2025

DBMS: Fundamentals & Advanced
Scaler · May 2025

Data Structures Demystified
Udemy · Nov 2024

Real-Time OS: Design & Impl.
Udemy · Nov 2024

Study Australia Industry Exp.
Practera · Jan 2023

EXPERIENCE

Web Developer Intern

CanIT Solutions · Canit Pulse (Multi-tenant Social Media Analytics SaaS) | Jun 2026 – Jul 2026
FastAPI · Supabase · React / TypeScript · Render

- Eliminated a production OOM crash on Render's free tier by architecting a path-selective raw SQL fetch layer, removing full in-memory loads of a ~19 MB JSON column.
- Closed a session-security gap caused by a token expiry bug (TOKEN_EXPIRE_HOURS not passed into token creation) — fixed creation path, added a missing client-detail API route, and implemented a global exception handler for structured error responses.
- Cut unnecessary frontend network load by resolving a re-fetch-on-tab-switch bug and JSX structural errors; shipped role-based permissions for report feedback and spec for a new Ad Spends module.

PROJECTS

SkillRoute — Proactive Capacity Engine

FAISS · FastAPI · React 19 · sentence-transformers

Flags stalling project tickets and routes them to the right internal expert via a three-part weighted Synergy Score (60% semantic skill match via FAISS, 25% calendar availability, 15% past success rate) and a break-even / cost-of-delay ROI calculator. Evolved from Streamlit prototype to production React 19 + FastAPI app. Full security audit fixed 12 vulnerabilities including a live API key committed to the repo, wildcard CORS, and missing auth/rate limiting — documented in SECURITY.md.

GitHub Innovator — Autonomous Agent

LangGraph · NVIDIA NIM · Typer · Textual

Fully autonomous six-stage LangGraph pipeline (discover → filter → LLM-analyze → rewrite → test → publish) producing forks 80–90%+ different from upstream with no human in the loop. Supports Python, JS, TS, Go, and Rust target repos. Ships with a CLI and interactive TUI dashboard for live monitoring and run history.

DeepFake Detection Platform

FastAPI · PyTorch · Docker · MinIO · PostgreSQL

9-service microservices platform detecting manipulated media across text, image, audio, and video. Orchestrator fans out uploads to independent detector services; a GradientBoosting fusion layer combines outputs into a single confidence score. Containerised end-to-end with Docker Compose, backed by MinIO object storage and PostgreSQL.

AI Ethics Advisor

React · Node.js · Ollama · Phi-3 Mini

Multi-framework ethical reasoning tool (Utilitarian, Deontological, Virtue Ethics) on a custom fine-tuned Ollama model (phi-3-mini-4k-instruct via LoRA). Delivers 100% valid JSON streaming with real-time token generation and AbortController-based stop. Hardened with bcrypt cost factor 12, 7-day JWT expiry, rate limiting, Helmet headers, and restricted CORS.

Dynamic Societal Friction Simulator (DSFS)

Python · MuRIL · GDELT · ACLED

Two-stage conflict forecasting pipeline — MuRIL-based encoder with trust learning and cleavage classification feeding a friction-tensor model with a Negative Binomial forecasting head. LTROC 0.73 at 4-week horizon, 12.3% above Hawkes-process baseline. IEEE paper draft. Co-authored through SRMIST CINTEL department.

SaaS Vending Machine

Terraform · Kubernetes · Ansible · AWS

Infrastructure-as-code system automating full multi-tenant SaaS provisioning across Terraform, Kubernetes, and Ansible. Deployable locally via Kind or on AWS (VPC, EKS, RDS, S3, IAM/IRSA). Security-hardened credential handling with dynamic password generation and an automated test suite.

Finance Tracker

FastAPI · React 18 · TypeScript · Chart.js

Full-stack personal finance app for budgeting, transaction management, and spending analytics. JWT-authenticated backend with PBKDF2+SHA256 password hashing; monthly, category-wise, and payment-method budgets with real-time tracking. Interactive dashboard with pie charts, calendar view, and a 30-item soft-delete recycle bin. Custom Tailwind design system with Framer Motion animations.

InsightX Analytics

Gemini 2.5 Flash · Streamlit · SQLite · Text-to-SQL

AI-powered Text-to-SQL engine for digital payment transaction analysis. Two-pass architecture: Gemini generates SQL from natural language, executes it against a real database, then a second pass produces insights strictly from returned data to prevent hallucination. Hardened with a read-only DB connection, SQL validation, rate limiting, and query caching.